Experiment 8

8.1 Implementation and Analysis of Decimation and Interpolation in Multirate Signal Processing Using MATLAB

Code:

clc; clear; close all; % Original Sequence n = 0:40; x = sin(0.3*pi*n); % Decimation by 2 y = x(1:2:end); nd = 0:length(y)-1; % Interpolation by 2 v = zeros(1,2*length(x)); v(1:2:end) = x; ni = 0:length(v)-1; % Figure 1: Original Sequence figure; stem(n,x,'filled'); grid on; title('Original Sequence x[n]'); xlabel('n'); ylabel('Amplitude'); % Figure 2: Decimated Sequence figure; stem(nd,y,'filled'); grid on; title('Decimated Sequence by 2'); xlabel('n'); ylabel('Amplitude'); % Figure 3: Interpolated Sequence figure; stem(ni,v,'filled'); grid on; title('Interpolated Sequence by 2'); xlabel('n'); ylabel('Amplitude'); % Figure 4: Comparison figure; stem(n,x,'b'); hold on; stem(0:2:length(v)-2,v(1:2:end),'r'); legend('Original x[n]','Interpolated Samples'); grid on; title('Comparison of Sequences'); xlabel('n'); ylabel('Amplitude');

Output:

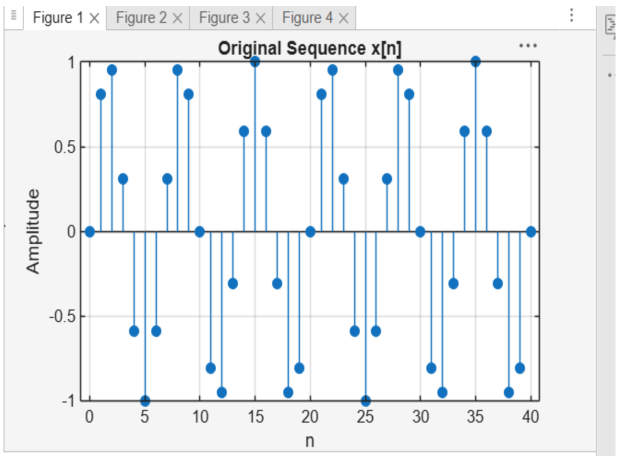
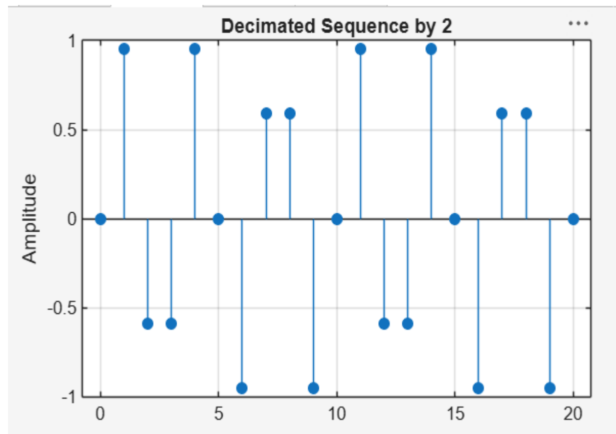
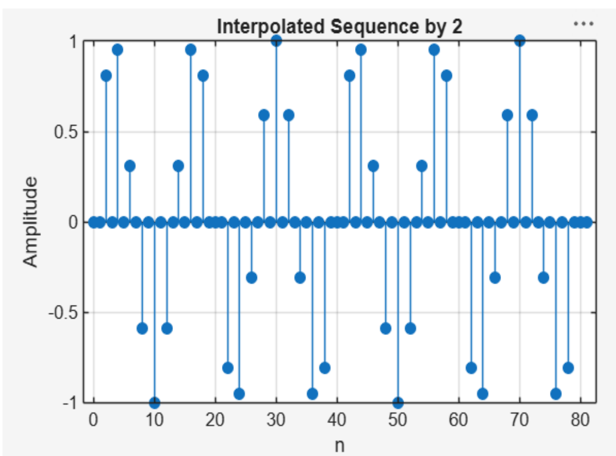


Figure 1 Figure 2 Figure 3 Figure 4
Original Sequence x[n]
Amplitude
n


Decimated Sequence by 2
Amplitude


Interpolated Sequence by 2
Amplitude
n

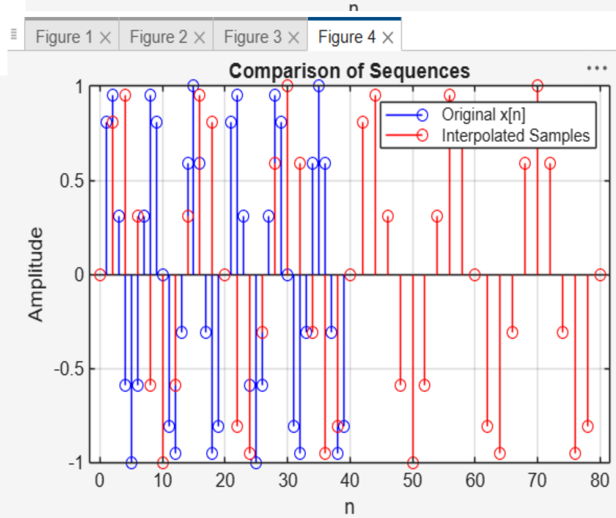


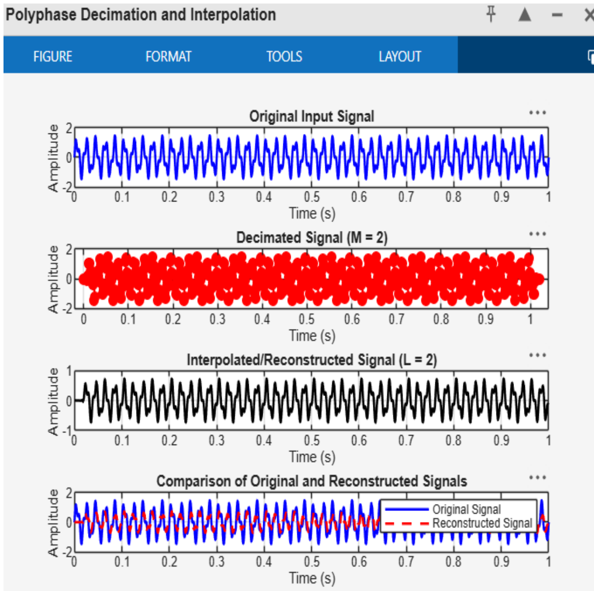
Figure 1 Figure 2 Figure 3 Figure 4
Comparison of Sequences
Original x[n]
Interpolated Samples
Amplitude
n

8.2 Polyphase Implementation of Decimation and Interpolation Filters Using MATLAB

Code:

```
clc; clear; close all; %% Input Signal Fs = 1000; % Sampling Frequency t = 0:1/Fs:1; % Test
Signal % Time Vector x = sin(2*pi*50*t) + 0.5*sin(2*pi*120*t); %% FIR Low-Pass Filter Design
N = 20; % Filter Order Fc = 0.4; % Normalized Cutoff Frequency h =
fir1(N,Fc,hamming(N+1)); %% Decimation Factor M = 2; % Polyphase Decimation y_dec =
upfirdn(x,h,1,M); %% Interpolation Factor L = 2; % Polyphase Interpolation y_int =
upfirdn(y_dec,h,L,1); %% Adjust Length for Comparison if length(y_int) > length(x) y_int =
y_int(1:length(x)); elseif length(y_int) < length(x) x = x(1:length(y_int)); end %% Time Axes t1 =
(0:length(x)-1)/Fs; t2 = (0:length(y_dec)-1)/(Fs/M); t3 = (0:length(y_int)-1)/Fs; %% BEST
GRAPH OUTPUT figure('Name','Polyphase Decimation and
Interpolation','NumberTitle','off'); % Original Signal subplot(4,1,1)
plot(t1,x,'b','LineWidth',1.5) title('Original Input Signal') xlabel('Time (s)') ylabel('Amplitude')
grid on % Decimated Signal subplot(4,1,2) stem(t2,y_dec,'r','filled') title('Decimated Signal
(M = 2)') xlabel('Time (s)') ylabel('Amplitude') grid on % Interpolated Signal subplot(4,1,3)
plot(t3,y_int,'k','LineWidth',1.5) title('Interpolated/Reconstructed Signal (L = 2)') xlabel('Time
(s)') ylabel('Amplitude') grid on % Comparison subplot(4,1,4) plot(t1,x,'b','LineWidth',1.5)
hold on plot(t3,y_int,'r--','LineWidth',1.5) title('Comparison of Original and Reconstructed
Signals') xlabel('Time (s)') ylabel('Amplitude') legend('Original Signal','Reconstructed
Signal') grid on %% Display Information disp('--------------------------------------');
disp('POLYPHASE DECIMATION & INTERPOLATION'); disp('--------------------------
```

Output:



Polyphase Decimation and Interpolation
FIGURE
FORMAT
TOOLS
LAYOUT
Original Input Signal
Decimated Signal (M = 2)
Interpolated/Reconstructed Signal (L = 2)
Comparison of Original and Reconstructed Signals
Original Signal
Reconstructed Signal
Amplitude
Time (s)

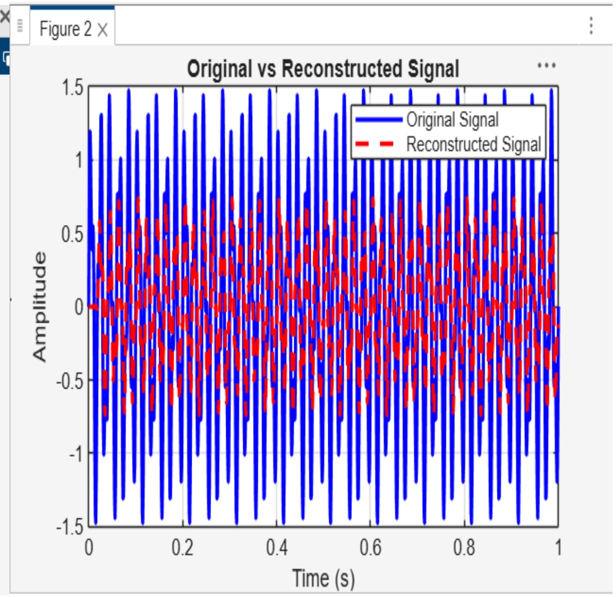


Figure 2
Original vs Reconstructed Signal
Original Signal
Reconstructed Signal
Amplitude
Time (s)